

Slingshot Stepsize Schedules for Conic-Particle Mean-Field Games: Experimental Report

NDDM Project

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Abstract

We test whether Shugart & Altschuler’s negative-stepsize “slingshot” schedules improve convergence of Chizat’s conic-particle algorithms (CP-MDA, $L = 1$; CP-MP, $L = 2$) to the Mixed Nash Equilibrium (MNE) of continuous two-player zero-sum games. Four games are studied: a bilinear game (Shugart’s own worked example), Chizat’s documented non-convergence case (a sine/cosine Fourier game on the torus, “Example 4.1”), and two well-conditioned convex-concave sanity games. The headline result is a split: slingshot cleanly rescues the bilinear game (duality gap collapses to machine precision vs. vanilla’s permanent plateau), but does *not* rescue Chizat’s Fourier game when particles start from a uniform spread over the whole domain. A warm-start ablation shows this is because the Fourier game’s failure is a *global* basin-of-attraction problem, not the *local* saddle-cycling problem slingshot’s theory addresses. This document explains the on-disk results layout (Section 2) and walks through every generated comparison panel (Sections 3–6). Section 8 closes by contrasting this toy-scale study with a proposed higher-stakes pivot, detailed separately in `References/preference_optimization_ideas.md`.

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1 How to read this report

Every panel below is a 2×2 figure produced by `plot_results.py` from the per-iteration CSV logs that `run_ablation.py` writes. In each panel:

- **Top-left:** Nikaido–Isoda duality gap (log scale) — the primary, domain-agnostic certificate of Nash-optimality. Lower is better; this is the metric to trust first.
- **Top-right:** summed Wasserstein distance from the current (μ, ν) to the registered `known_mne` (log scale), where one is registered. Co-primary in general, but **unsound specifically for the bilinear game** (Section 3.3).
- **Bottom-left:** entropy $H(a)$ of the particle weights — a secondary diagnostic of how concentrated the Fisher–Rao weights have become.
- **Bottom-right:** $\|\text{barycenter}\|$ (log scale), i.e. $\|(\mathbb{E}_\mu[x], \mathbb{E}_\nu[y])\|$ — secondary, but the decisive metric for the bilinear game specifically.

Color convention throughout: vanilla baseline is purple, slingshot is red. All seeds for a given configuration are overlaid in the same color.

2 Results directory layout

The `results/` tree is partitioned per game, and further split into sub-regimes where a single game admits more than one meaningful experimental setup. Every leaf directory contains: one CSV per run (`L{1,2}_{vanilla,slingshot}_init-{uniform,near_mne}_seed{N}.csv`), one `summary.csv` aggregating final metrics and fitted log-linear rates across all runs in that directory, and a `plots/` subfolder holding the PNG dashboards discussed in this report. A single `summary_all.csv` at the `results/` root concatenates every summary row across every game.

2.1 results/bilinear/

Shugart’s own worked example, $f(x, y) = xy$ on a bounded euclidean domain.

`pitfall_default_stepsize/` The run at the *default* stepsize calibration. Kept deliberately, with an explanatory `README.md` in the same folder, as a documented negative result: bilinear’s spectrum is degenerate ($m = M = 1$), so both the vanilla stepsize ($h = 1/\sqrt{M} = 1$) and the Chebyshev/slingshot schedule (which collapses to the same constant magnitude when $m = M$) pick step size 1.0 — far above the position-step’s adaptive clip cap. Both modes saturate the clip and freeze at the domain boundary within a handful of steps, making vanilla and slingshot look identical for the wrong reason.

`calibrated/` The same comparison re-run with a deliberately small step ($h = 0.01$, via `--eta 0.01` for vanilla and `--m 10000 --M 10000` for slingshot) to stay below clip saturation. This

is the run that actually isolates slingshot’s effect, and produces the cleanest positive result in this study.

2.2 results/chizat_example_4_1/

Chizat’s documented non-convergence case: a sine/cosine Fourier-spectrum payoff on the torus $[0, 2\pi) \times [0, 2\pi)$.

global/ Particles initialized uniformly at random over the whole torus — Chizat’s actual experimental setup. This is the regime in which Chizat reports CP-MDA fails to converge.

local/ Particles warm-started with small Gaussian noise around the atoms of the known MNE (`init_state_near_mne`, noise $\sigma = 0.01$). This regime isolates whether the failure is *local* cycling near the saddle (which slingshot’s theory is proven to fix) or something else entirely.

2.3 results/convex_concave/

A smooth, strongly-monotone convex-concave sanity game ($f(x, y) \propto \tanh$ -type one-sided gradients) on a euclidean domain. No sub-split: standard vanilla-init runs across both $L = 1$ and $L = 2$.

2.4 results/sc_sc/

A second, better-conditioned sanity game (“sin-cos” on a euclidean domain), included to check that nothing about the implementation itself prevents fast, clean convergence when the underlying game is easy. No sub-split.

3 Bilinear game

3.1 Pitfall: default stepsize calibration

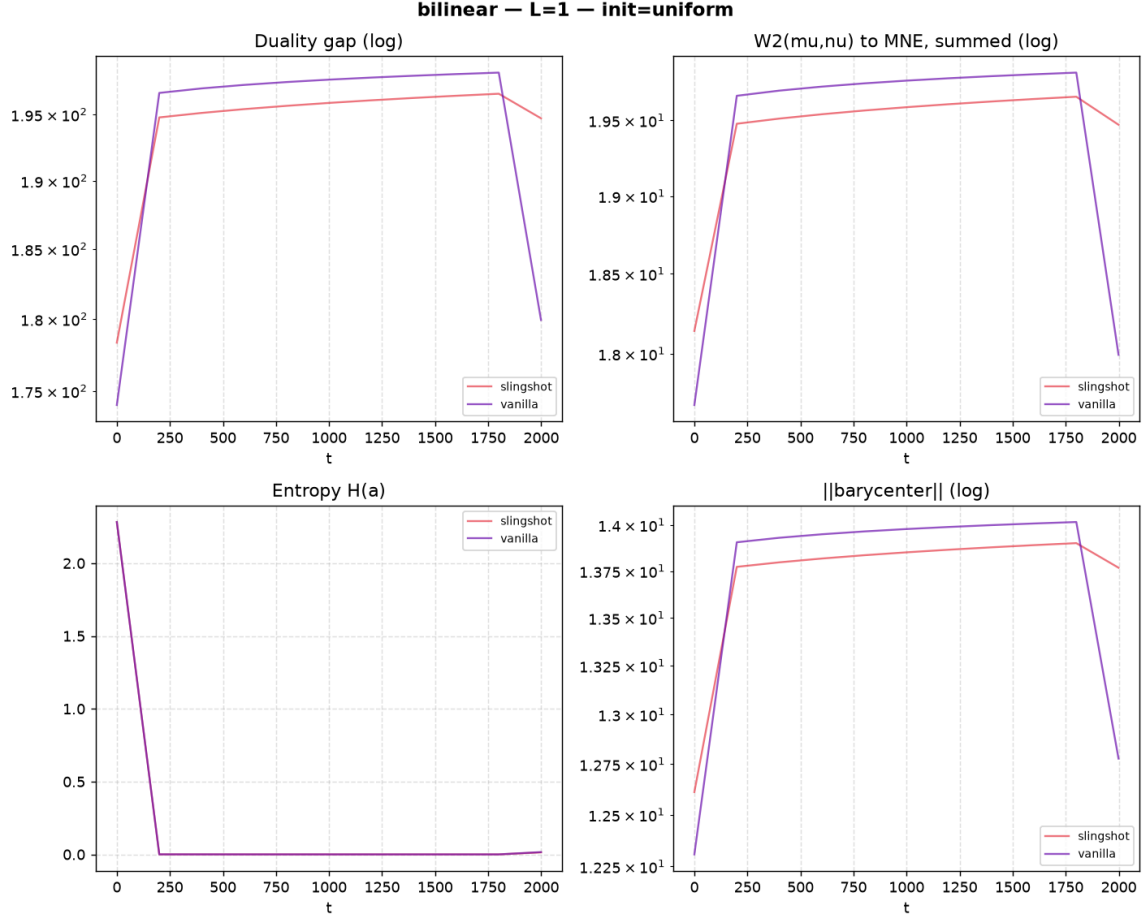


Figure 1: Bilinear, $L = 1$, default stepsize. Vanilla and slingshot are visually indistinguishable: both saturate the position-step’s adaptive clip within a few iterations and freeze at the domain boundary, so the duality gap plateaus near 190–200 and entropy collapses almost immediately for *both* modes. This is a calibration artifact of bilinear’s degenerate spectrum ($m = M = 1$), not a finding about the sign schedule — see [results/bilinear/pitfall_default_stepsize/README.md](#).

3.2 Calibrated comparison

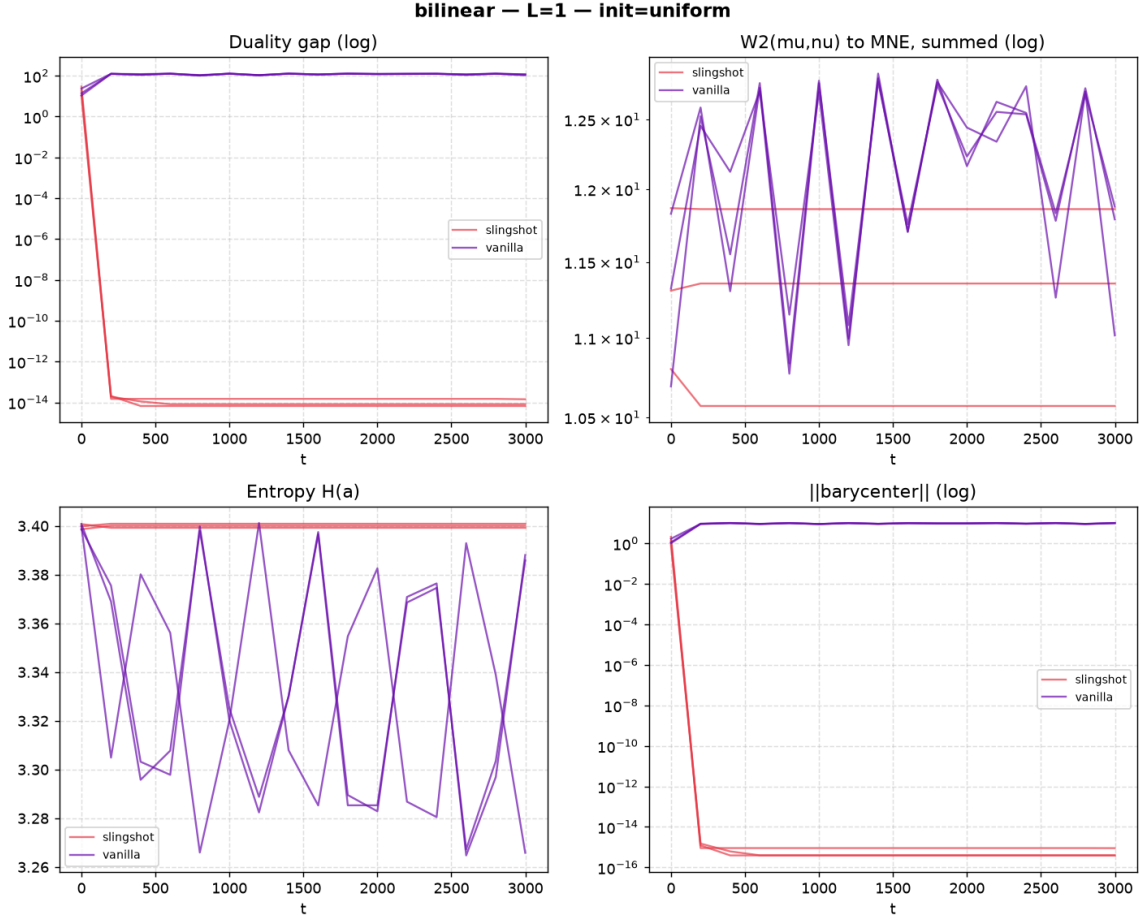


Figure 2: Bilinear, $L = 1$, calibrated stepsize ($h = 0.01$, below clip saturation). Slingshot’s duality gap (red) collapses to machine precision ($\sim 10^{-14}$) and stays there; vanilla (purple) stays stuck at ~ 106 – 117 across all seeds. The barycenter-norm panel tells the same story directly: slingshot’s barycenter goes to $\sim 10^{-16}$ while vanilla’s barycenter sits far from zero. This is the cleanest reproduction of Shugart’s headline claim in this codebase.

3.3 Caveat: the Wasserstein-to-MNE panel is not meaningful here

In both bilinear panels above, the top-right (W2-to-MNE) curve stays roughly flat for *both* modes, which at first looks like a contradiction of the duality-gap result. It is not a bug. Under product measures, the bilinear payoff is separable:

$$F(\mu, \nu) = \mathbb{E}_{x \sim \mu}[x] \cdot \mathbb{E}_{y \sim \nu}[y],$$

so the duality gap depends *only* on the two barycenters, never on the shape or support of μ, ν individually. The true MNE set is therefore the entire family of zero-mean (μ, ν) pairs, not just the single point-mass-at-zero representative registered as `known_mne`. Slingshot’s particles spread out across the domain (individual positions ranging over roughly $[-10, 9.6]$ in the runs here) while their *barycenter* converges exactly to zero — so W_2 to the canonical representative stays large even though the game is solved. **Conclusion: trust the duality gap (and the barycenter panel)**

as primary for bilinear; the W_2 panel only measures convergence to one arbitrary representative of the true MNE set, not solution quality.

4 Chizat’s Fourier game (Example 4.1)

4.1 Global initialization (Chizat’s actual setup)

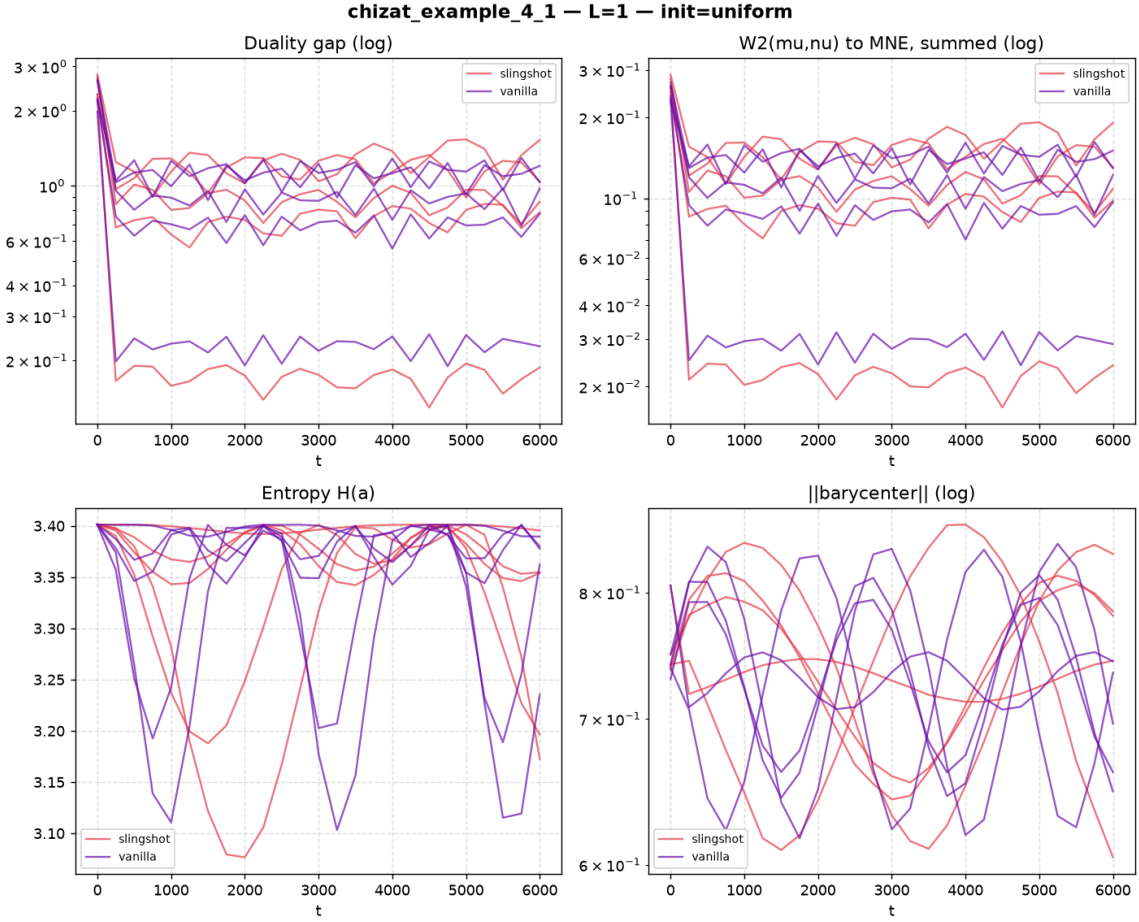


Figure 3: Fourier game, $L = 1$ (CP-MDA), uniform random initialization over the whole torus. Vanilla and slingshot overlap almost exactly: the duality gap oscillates in a persistent band with no decay trend, and the entropy dips but never collapses — both modes are cycling/plateauing, not converging. This reproduces Chizat’s documented non-convergence claim, and slingshot provides no visible rescue.

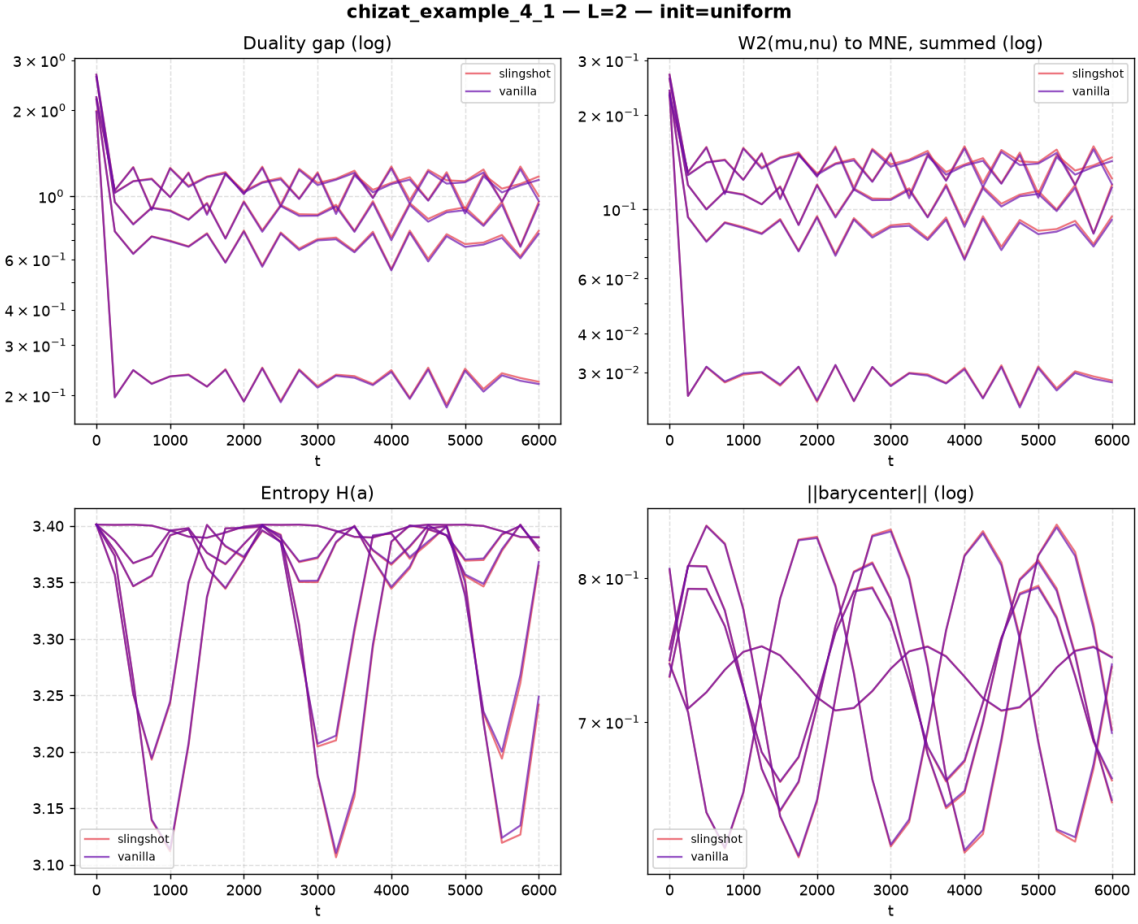


Figure 4: Same setup with $L = 2$ (CP-MP / extragradient). The extragradient correction does not change the qualitative picture either: both modes still plateau without separation.

4.2 Local initialization (warm-started near the known MNE)

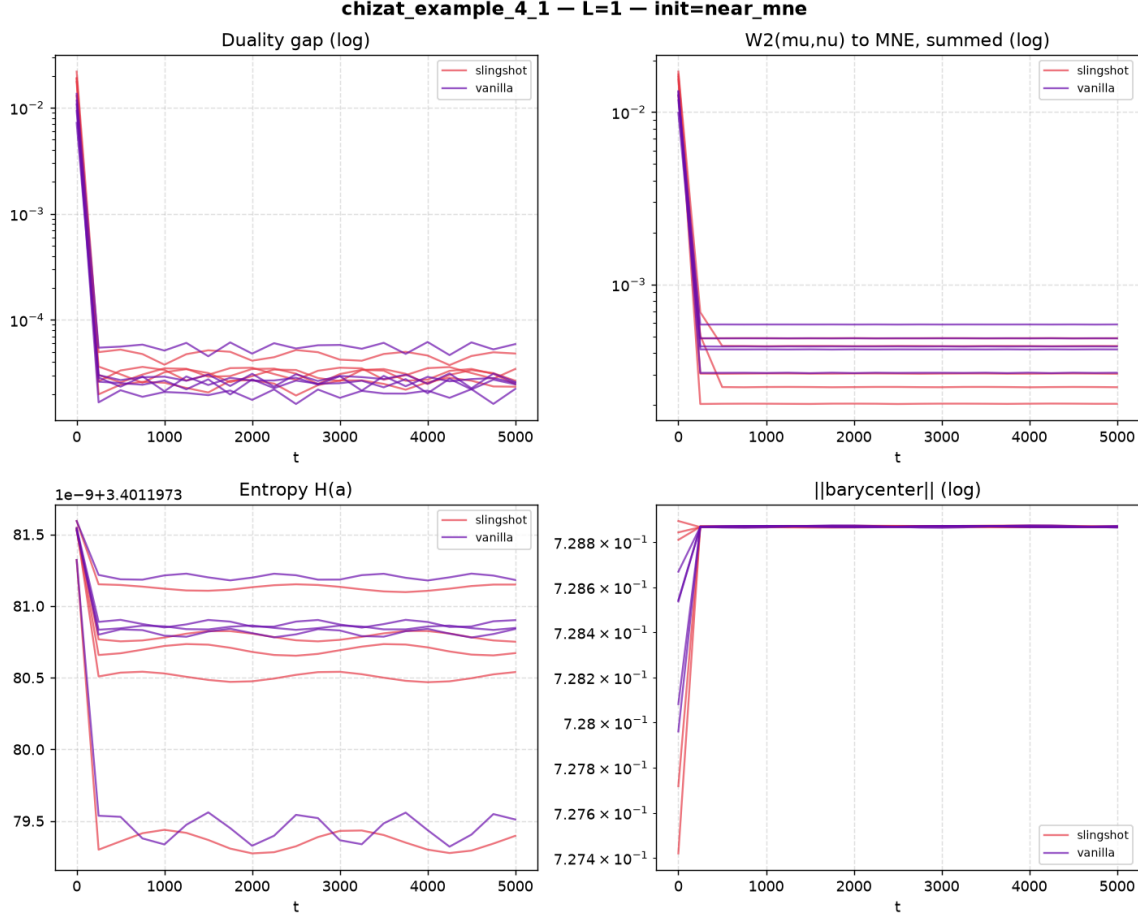


Figure 5: Fourier game, $L = 1$, particles warm-started with small noise ($\sigma = 0.01$) around the true MNE support. Both modes collapse almost immediately to a tiny residual duality gap ($\sim 2.5 \times 10^{-5}$, confirmed not a grid-discretization artifact by comparing `grid_size= 400` vs. 2000) and hold there for the remainder of the run. Entropy stays pinned essentially exactly at $\ln(30) \approx 3.4012$ throughout (only ninth-decimal-place drift), meaning the Fisher–Rao weights never needed to move away from the warm-started uniform $1/n$ initialization — all the necessary information was already encoded in particle *positions*.

4.3 Reading: a global basin-of-attraction failure, not local cycling

The contrast between the two panels above is the central finding of this study regarding the Fourier game. Shugart & Altschuler’s slingshot guarantee is a statement about *local* stability near a saddle: alternating the sign of the stepsize prevents the kind of small-scale cycling that traps vanilla gradient descent-ascent near an equilibrium. The local-init panel shows that once particles are already *near* the right support, neither mode has any local cycling problem to fix — they both converge fine, trivially. The global-init panel shows the real failure mode: starting from a uniform spread over the whole torus, the Fisher–Rao weight dynamics never manages to concentrate mass onto the correct support in the first place. That is a basin-of-attraction / exploration problem, and slingshot’s mechanism (a local position-step sign flip) has no obvious lever on it. This explains why slingshot

does not rescue Chizat's documented failure case: the failure documented there is not the failure mode slingshot's theory was built to fix.

5 Convex-concave sanity game

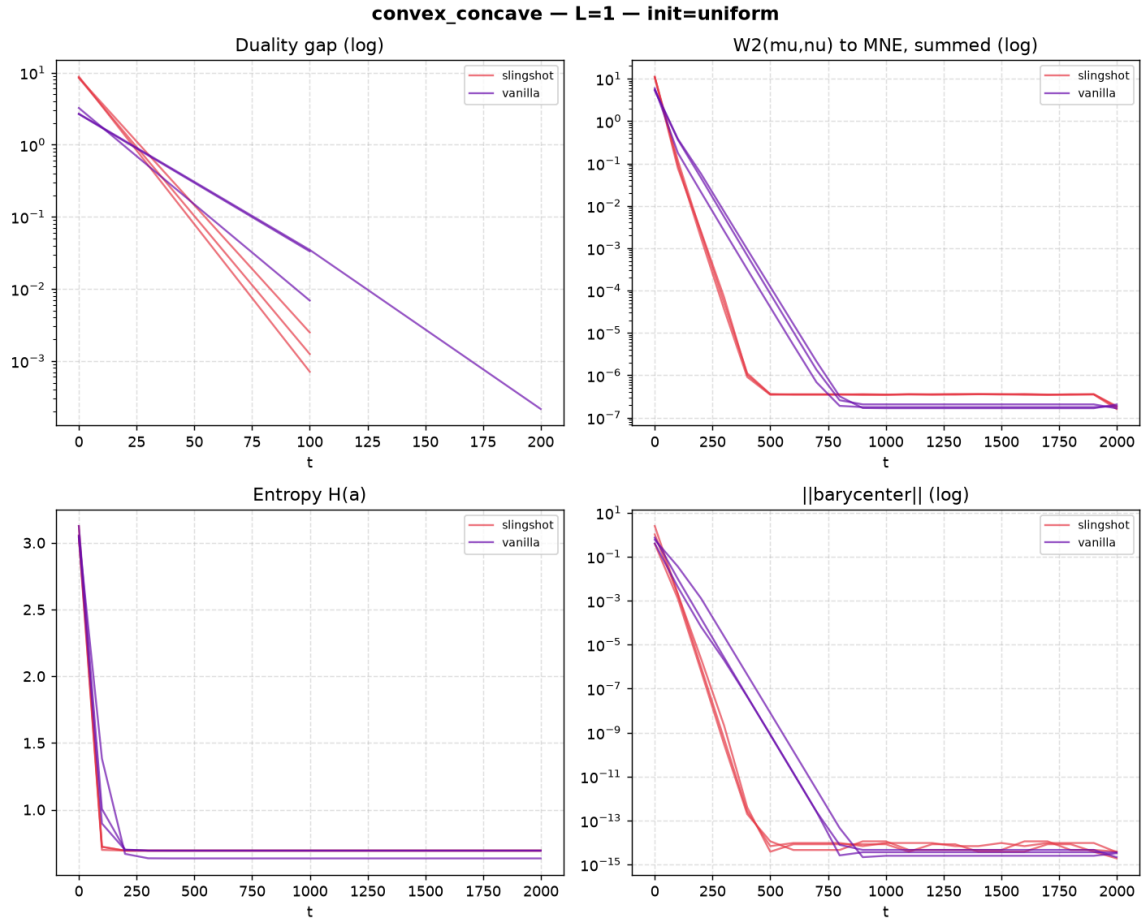


Figure 6: Convex-concave game, $L = 1$. Both modes converge, with broadly similar trajectories.

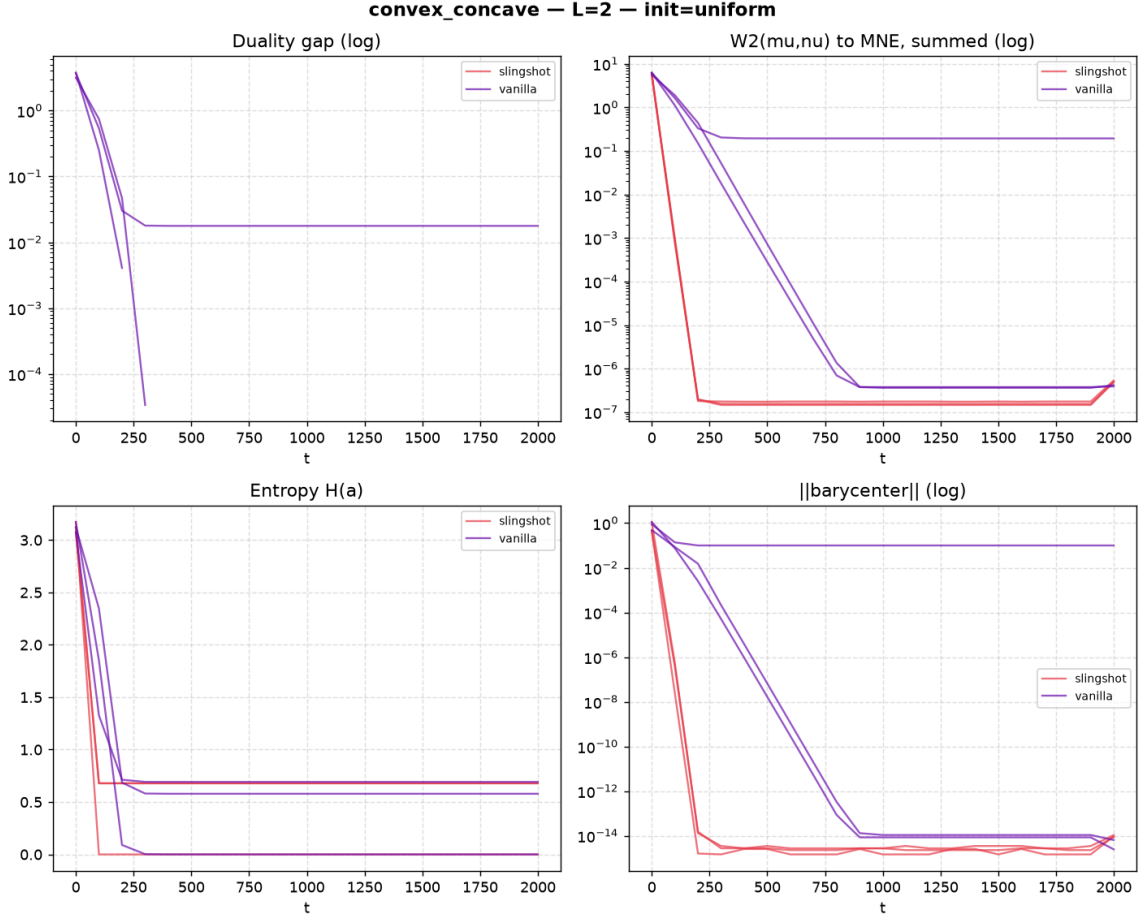


Figure 7: Convex-concave game, $L = 2$. Clean, monotone convergence for both modes. This is an easy, well-conditioned game by design: it is not the regime where vanilla GDA is expected to fail, so the absence of a large slingshot advantage here is consistent with theory, not a contradiction of the bilinear result.

6 Sin-cos sanity game

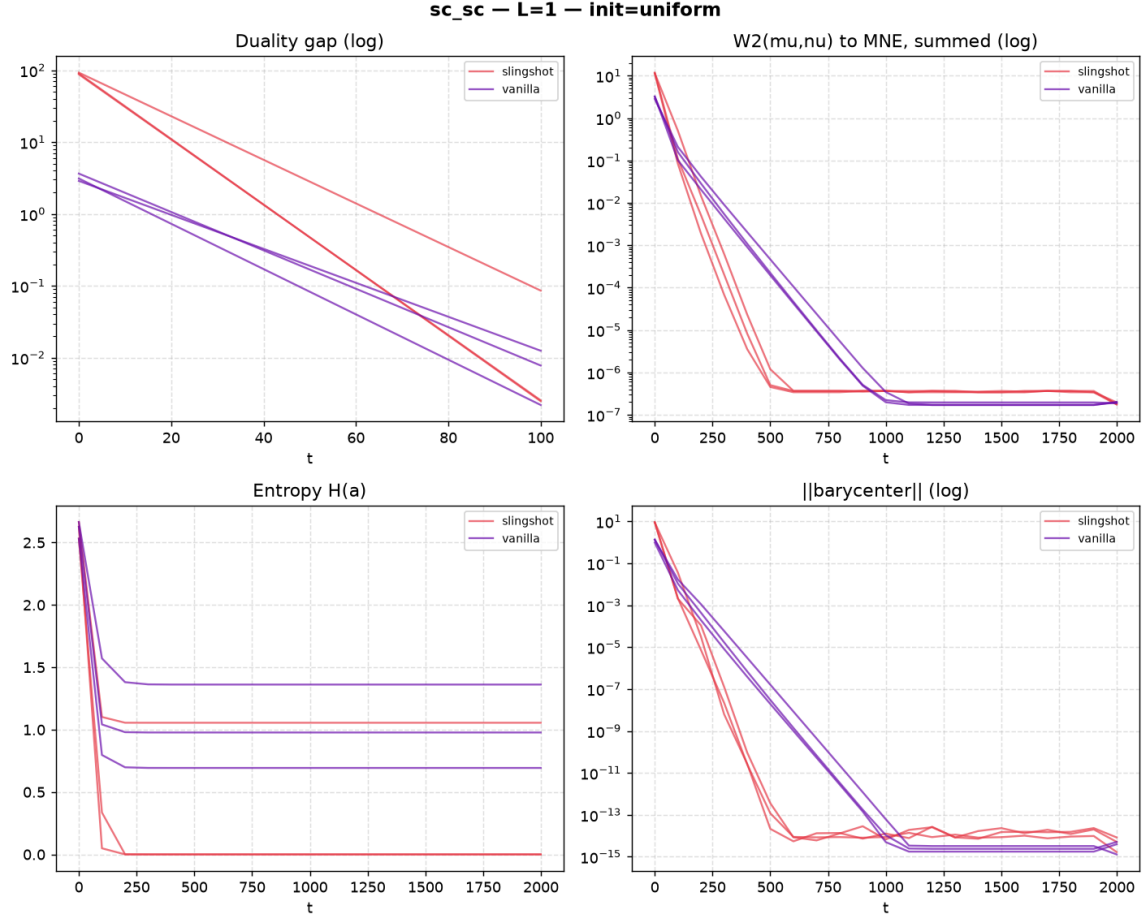


Figure 8: Sin-cos game, $L = 1$. Both modes converge very fast. The duality-gap panel’s visible x -range truncates around $t \approx 100$ – 150 : once the gap drops below the plotting floor, `plot_results.py`’s `_semilogy_safe` helper masks non-positive/near-zero values before taking the log, so the curve simply stops being drawn rather than diving to $-\infty$. This is a plotting-floor artifact, not a sign that the run stopped early — it reflects genuinely fast convergence on the best-conditioned game in the suite.

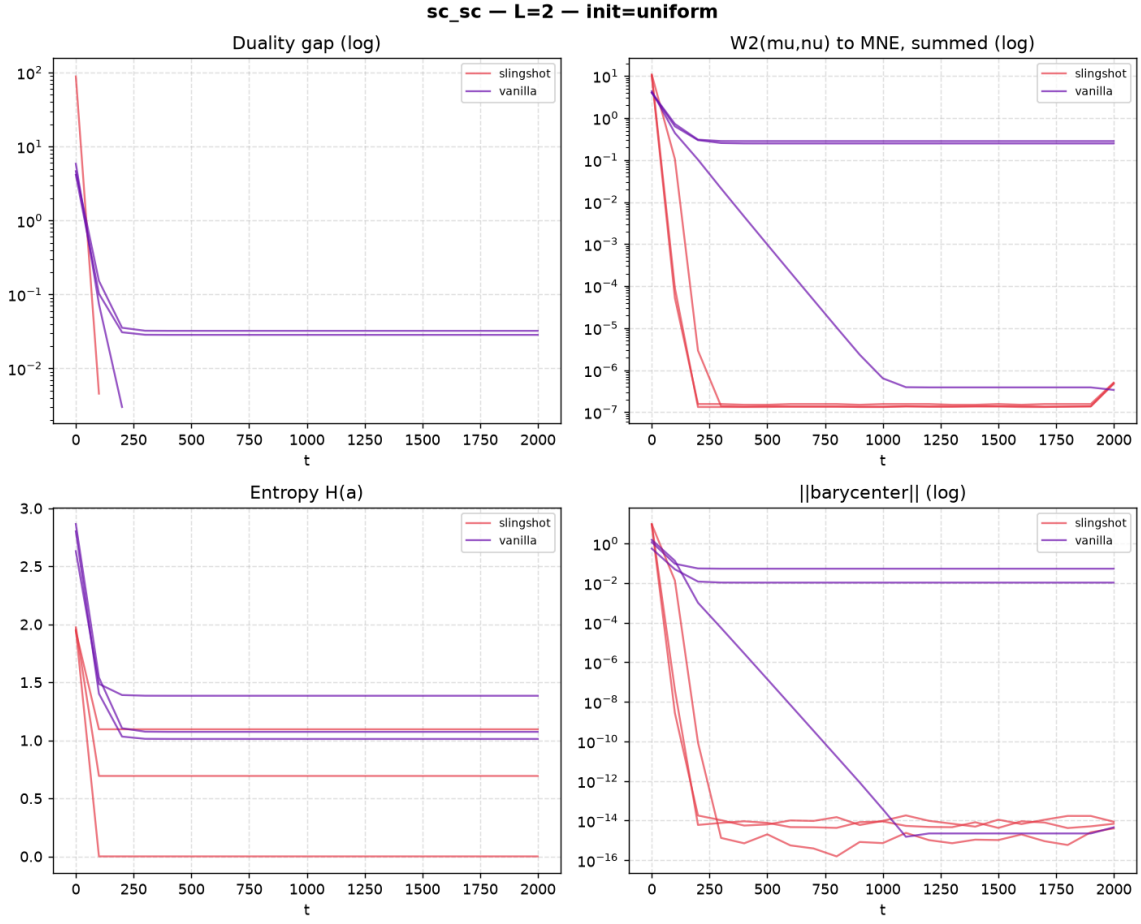


Figure 9: Sin-cos game, $L = 2$. Same fast-convergence picture as $L = 1$.

7 Summary of findings

Bilinear Clean positive result for slingshot once calibrated below clip saturation: duality gap to machine precision vs. vanilla’s permanent plateau (Section 3).

Fourier (Chizat 4.1), global init No rescue: vanilla and slingshot are indistinguishable, both plateauing/cycling, reproducing Chizat’s documented non-convergence claim (Section 4).

Fourier (Chizat 4.1), local init Both modes converge trivially when warm-started near the true MNE, showing the global-init failure is a basin-of-attraction problem outside the scope of slingshot’s local-stability guarantee (Section 4).

Convex-concave, sin-cos Both sanity games converge cleanly under both modes, as expected for well-conditioned games that are not slingshot’s target failure case (Sections 5–6).

Full implementation notes, bug-fix history, and the reasoning behind each experimental design choice are recorded in `References/CLAUDE.md` at the project root.

8 Outlook: from toy games to a real application

Taken alone, the study above is a useful but limited piece of work: one clean positive result on a textbook bilinear example, one carefully diagnosed non-result on the case that actually motivated the investigation, and no new theory. Table 1 makes the point concretely by contrasting this study with a proposed pivot, written up in full in `References/preference_optimization_ideas.md`.

Table 1: This toy-game study vs. the proposed preference-optimization pivot.

	This study	Proposed pivot
Problem scale	1–2 dimensional toy games (bilinear, torus Fourier, convex-concave)	Nash-equilibrium-seeking preference optimization for LLM alignment
Stakes	Tests an existing theory paper’s claims on hand-picked examples	One of the most active subfields in ML right now (Nash-MD, SPPO, DNO, INPO, Nash-Prox, Magnetic Mirror Descent are all 2024–2025 work)
What’s novel	Re-implementation, bug-fixing, and diagnosis of <i>why</i> an existing claim does or doesn’t hold	Slingshot has never been tried against this literature’s cycling problem — confirmed by targeted search, June 2026
Headline result	Split: positive on bilinear, explained non-result on Chizat’s case	Open question, with falsifiable sub-hypotheses and a staged, mostly-code-reuse experimental plan
Path forward	Workshop note / negative-results note	Full paper: cheap synthetic-game test (days, reuses this repo’s machinery) → small-LM scale-up → theory extension

The connection is closer than it looks. Nash-MD, SPPO, and INPO are all, underneath their RLHF framing, running mirror descent or multiplicative weights on a two-player constant-sum game in search of its Nash equilibrium — exactly the object this codebase’s `algo/conic_particle.py` Fisher–Rao weight update already computes, and exactly the dynamical family (GDA / mirror descent on games) that cycles, and that slingshot’s negative-stepsizes schedule was built to fix. The existing `slingshot/` schedules and `run_ablation.py` pipeline carry over with comparatively little new code: the first proposed experiment is a finite intransitive preference matrix game (a direct, fixed-support, position-step-disabled instance of the same WFR machinery used throughout this report) under vanilla self-play vs. slingshot, mirroring the bilinear-vs-Fourier comparison above but on a problem with real stakes. See `References/preference_optimization_ideas.md` for the full literature landscape, the staged experimental plan, and the central open risk (whether slingshot’s second-order cancellation mechanism survives the stochastic gradients a real preference-learning setting introduces).